TRANS-ANATOLIAN NATURAL GAS PIPELINE (TANAP) - 56" MAINLINE ITP

Pipe Description: 56" (1422 mm) x 19.45 mm, Grade X70M PSL 2, SAWH Pipe | **Governing Standard:** TANAP Quality Specification / API Spec 5L 47th Edition PSL 2

No	Inspection / Test Activity	Frequency & Extent	Sampling Location & Specimen	Test Method / Standard	Acceptance Criteria & Specified Limits	Governing Clause	Hold / Witness
1	**Ladle Heat Chemical Analysis**	One analysis per heat of steel	Ladle sample	ASTM A751	C <= 0.10%, P <= 0.015%, S <= 0.004%, Nb+V+Ti <= 0.15%, CE_Pcm <= 0.20	TANAP Spec / API 5L	H/W
2	**Product Chemical Analysis**	Two analyses per test unit (lot <= 100 pipes)	Pipe body coupon	ISO 14284	C <= 0.10%, P <= 0.015%, S <= 0.004%, CE_Pcm <= 0.20	API 5L Tablo 5	W
3	**Pipe Body Tensile Test**	Once per test unit	Transverse body coupon	ASTM A370	Rt0.5: 485-635 MPa, Rm: 570-760 MPa, Af >= 19.0%, Y/T <= 0.90	API 5L Tablo 7	H/W
4	**Weld Seam Tensile Test**	Once per test unit	Transverse weld coupon	ASTM A370	Rm >= 570 MPa across helical weld seam	API 5L Tablo 7	W
5	**CVN Charpy Impact Test (-20 °C)**	1 set body + 1 set weld at -20 °C	Transverse Charpy V	ASTM A370	At -20 °C: Min Average 80 Joules, Min Individual 60 Joules	TANAP Spec	H/W
6	**DWTT (Dü■en A■■rl■k Y■rt■lma Testi) at 0 °C**	Once per heat at 0 °C	Transverse DWTT coupon	API RP 5L3	At 0 °C: Average shear area >= 85%, No single specimen < 60%	API 5L Tablo 18	H/W
7	**Guided-Bend Test**	1 root + 1 face bend per test unit	Transverse weld coupon	ASTM A370	No defect or opening > 3.2 mm after 180° bend	API 5L 9.10	W
8	**Hardness Testing (HV10)**	Once per test unit	Cross section macro	ASTM E384	Maximum 280 HV10 across Body, HAZ and Weld	TANAP Spec	W
9	**Residual Stress Ring Test**	Every heat (per heat mandatory)	150 mm ring opposite weld	TANAP Spec	S <= 0.10 x SMYS (Max 48.5 MPa)	TANAP Spec	H/W
10	**Mill Hydrostatic Pressure Test**	Each pipe (100% all 56" pipes)	Full pipe length	API 5L 10.2.6	Test Pressure: 100% SMYS (134 bar), Duration: MINIMUM 10 SECONDS	API 5L 10.2.6.2	H/W
11	**Weld Seam 100% Automated UT + RT**	100% full length of helical weld seam	Full weld seam	ISO 10893-11/6	100% Online UT + Real-time Radioscopy of weld seam ends & repairs	API 5L Ek E	H/W
12	**Strip Body & Pipe Ends UT Laminar Scan**	100% strip scan and 100% pipe ends	Full strip & pipe ends	ISO 10893-9/8	100% strip UT (Class E2); 100 mm pipe ends (no defect > 6.0 mm)	API 5L Ek E	W
13	**Dimensional, Straightness & Marking**	100% automated laser measuring station	Pipe body & ends	TANAP Spec	Diameter: +-0.50%; End ovality <= 50% Tablo 10; Straightness <= 0.10% L	TANAP Spec	W
14	**Visual Surface & Coating Surface Prep**	100% internal and external surface	Full pipe surface	API 5L 10.2.7	100% visual inspection; surface cleanliness ISO 8501-1 Sa 2.5 for 3LPE	TANAP Spec	W